



How to crash: flight manual

The Circus.

Ernesto De Felice - Pierluca De Felice - Raffaele Colamarino - Raffaele Aucelli

1. Prologue

«Everyone must do their bit.»

But what is mine?

I spent years chasing a star I could never quite see, and in all honesty, I never imagined my path could smell of... What on earth is that smell? Candy floss?

I had only just earned my pilot's licence when they arrived; had they shown up a little earlier, I would have saved myself no end of bother. The crowd was milling around the enormous screen in the centre of the square and I, in all honesty, hardly paid any attention: I was used to wandering off in my own thoughts without truly watching where I was heading, or who was around me.

"No bloody way!" someone shouted from the middle, or something rather more foul-mouthed; my earphones and the slurp of my fizzy drink kept me from hearing properly. I turned towards the screen, expecting yet another missed goal by some footballer. How wrong I was.

I had grown up on sci-fi tales of colossal, technologically advanced starships capable of ploughing through the dark, infinite spacetime tide in a humanly reasonable amount of time. Instead, onto that screen flashed footage from a satellite orbiting the Moon to capture high-resolution video of our beloved neighbour: the regolith, the craters, the LEM, a circus tent... **A CIRCUS TENT?!**

2. Interview No. 339

Remember to smile at the applicants.

Interviewer: «Everyone must do their bit.»

Pilot: «I haven't quite grasped what mine is supposed to be.»

Interviewer: «We have a... what's it called again?»

Clown: «Giovanni.» *(He was sitting next to my interviewer, staring right at me.)*

Interviewer: «No, I meant the enormous mothership cannon threatening to obliterate Earth unless we manage to pack out all the circus tents they've pitched across the Solar System.»

Clown: «Yes, Giovanni.» *(He crossed one leg over the other, then swapped them back, as if uncomfortable, maintaining that infernal grin.)*

Pilot: «Hang on, so it's actually true?»

Interviewer: «An unknown alien species is threatening our planet; their population, currently on holiday, will be stopping by the various big tents pitched across celestial bodies. They will spare us only if the audience laughs at every single show.»

Pilot: «So my destiny is being decided by a board of directors wearing red noses?!»

The clown locked eyes with him and, at last, with a swift squeeze of his bulbous red nose, produced a loud, playful honk: **HONK!**

3. Mission Planning

Clown: «Start wherever you fancy, mind you; we have to fill every big tent on every celestial body with a solid surface: wouldn't want to go taking an accidental dip, would we? HAHA! Honk!»

Pilot: «This is supposed to be my first trial day...»

Clown: «Fair enough, if it's your first time, kick off with the Moon. If you come back alive, you can pick the other destinations. Onboard you'll be with Nasino; she's a bit of a cheeky AI, I'd recommend not making her angry.»

3.1 Game Mode: Simulation or Fast

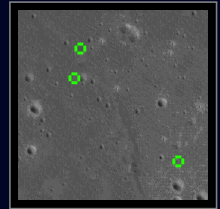
Clown: «Before climbing aboard, decide how you'd like to get hur- that is, how you want to proceed: slow and steady, or do you want to get it over with quickly? Welcome to the Mode selection! We've also catered for those with little patience, and in our species there are quite a few who share this trait...»

- **Simulation:** The original, realistic flight experience. RCS thrust and rotation speed follow stricter rules. The experience tends to last longer in this Mode; certainly longer than your nerves.
- **Fast:** Circus engineers tampered with the RCS thrusters, pushing them well past their safety limits and boosting their thrust by **A WHOPPING TEN TIMES (10x)**. Module rotations are waaaay faster, drastically reducing the time required. Brilliant if you want to be quick, terrible if you have heavy fingers!

3.2 Selecting the celestial body

Nasino: «Hello! I'm Nasino!! Select your target here - djak,c°@38 - My mistake. If it's your first time... CONGRATULATIONS! You will only be able to choose the Moon. Keep a sharp eye on fundamental parameters:»

- **"Starting":** Altitude below which you must take manual control, because there's not a chance in hell I'm doing everything myself!
- **"Gravity":** How rapidly you are drawn towards your demise. A higher value means your thrusters will have to work much harder.
- **Description / Reviews:** Handy for understanding what mess you're getting into (customer reviews? I shan't be showing you those).
- **Flight Mode (Day / Night):**
 - *Day Mode:* Descent under standard solar illumination; terrain and touchdown surface fully visible.
 - *Night Mode:* Pitch black, I mean what on earth were you thinking? The surrounding environment is cloaked in darkness, and you will rely solely on the onboard spotlight to spot the surface and proper landing site.



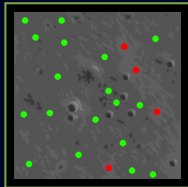
Target: Moon

3.3 Choosing the landing area

Nasino: «Right, you can't just go landing on the big tent, that wouldn't be very polite! For each Celestial Body, three landing sites are planned. However, owing to bureaucratic delays, **currently only the first area is operational!** Don't roll your eyes at me!!!

I hope you're familiar with the International System of Units, we'll only be using that one!

... What? The Clap Index isn't part of the SI? They really should add it...»



Area 1: example of the landing zone with pads (highlighted) and without.



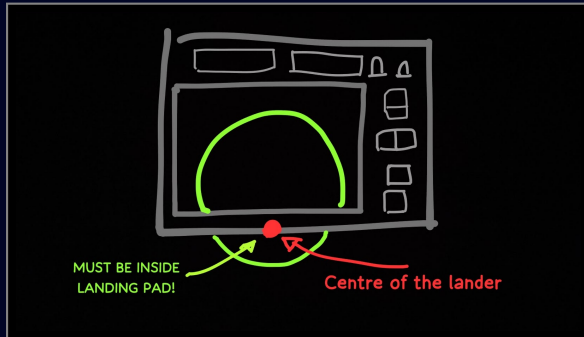
Areas 2 & 3 (COMING SOON...)

Works in progress: clearing boulders.

Once all three are active, each will feature a distinct topography, its own difficulty rating ("Diff"), and appropriate score multipliers.

3.4 How to land properly

Nasino: «Oh, I almost forgot! To land properly, the centre of the lander must be inside the landing pad! So, considering that the camera displaying telemetry on your screen is mounted on one side, you should have a situation like this:»



«What do you want? I'm not a generative AI, make do with this example.»

4. Flight Configuration and Resources

Nasino: «It's not you we need; or rather, we only need you to deliver your crew members down in one piece so they can make the aliens chuckle... For now, your selection is restricted to a single crewmate alongside yourself, seeing as you're on probation. Let's minimise collateral damage, nothing personal.»

4.1 Selecting the lander

What will you use to complete your mission? The choice is yours.

- **"Crew":** How much fresh meat you can haul. Increasing the headcount increases total mass, with direct consequences on fuel consumption.
- **"Mass" & "Fuel":** Total dry mass to transport and maximum propellant capacity available. Watch it closely!
- **"Thrust" & "RCS":** Maximum thrust of primary engines and auxiliary thrusters for rotation and translation (do try landing lopsided and let me know how that pans out).

4.2 Resource selection (Critical Stage)

WARNING: The Circus accepts no liability for loss of human life. The pilot bears sole responsibility. Think of it like a trip to the butcher's: with "Crew" you are choosing your cuts of meat. The more you pick, the heavier you get, and you already know the drill for the rest. Perhaps one day we will assign a different Clap Index modifier for each individual crew member. The applause, the applause is what pays your rent.

5. Cockpit Controls and Flight Deck

5.1 Flight & Manoeuvring Controls (Cockpit) - How to avoid crashing into bits.

- **A: Main Engine** - Fires the primary vertical thruster to counteract gravity. Consumes propellant.
- **D-PAD Left / Right: Lateral RCS** - Delivers horizontal manoeuvring thrust to the left or right.
- **D-PAD Up / Down: Vertical RCS** - Delivers translational thrust upwards or downwards.
- **L / R (Shoulders): LEM Rotation** - Clockwise (L) or counter-clockwise (R) roll attitude.
- **SELECT:** Popipopi. Press it. Go on, do it.
- **START:** Flight pause: pause is pause.



Cockpit Telemetry Display

5.2 Auxiliary Systems: Radar and Spotlight

- **B + R: Onboard Radar** - Toggles radar scanning for the touchdown pad.
CAUTION: The radar engages as soon as you activate it, both inside and outside the landing zone, but only if you are below 200 metres of altitude.
- **B + L: Landing Lights (Spotlight)** - Toggles the high-intensity searchlight illuminating the ground below.
- **Warning Light:** When you are outside the landing zone and below the descent procedure starting altitude (200 m), the warning light will illuminate on your screen. Don't get distracted, it's very large.

Regarding energy consumption:

The landing light functions **ONLY and EXCLUSIVELY in Night Mode!**

If you are in Day Mode and attempt to turn it on, the light **will NOT illuminate** (seeing as the sun is already out, don't be greedy), **BUT YOU WILL STILL CONSUME 1 UNIT OF BATTERY POWER!**

Avoid pressing buttons at random; you're supposed to be a professional.

5.3 Menu Navigation Controls

In selection, configuration, and pause menus:

- **A: Confirm / Advance** - Selects highlighted option, opens technical dossiers, launches flight, or confirms Game Mode (Simulation/Fast), landing area, or lander.
- **B: Back / Cancel** - Returns to the previous screen or cancels the last selection.
- **D-PAD Up / Down: Cursor Movement** - Selects Game Mode (Simulation / Fast), navigates vertically through landing areas and pause menu entries.
- **D-PAD Left / Right: Value Adjustment** - Toggles Flight Mode (Day Mode / Night Mode), cycles through celestial bodies and landers, adjusts crew headcount.
- **START: Pause** - Opens the pause menu during flight.

Nasino: *«Press Select, go on, I dare you»*